

CarePulse — Modern UI + Android Features Enhancement Plan

Context

CarePulse is a native Android (Kotlin + Jetpack Compose, Material 3, Firebase) caregiver app currently on branch `feature/firebase-bottom-nav-polish`. The Navy + Teal rebrand, bottom nav, real-time chat, and Pulse dashboard are in place. The user wants the app to feel **modern**: refreshed button colors, richer animations, deeper Android-platform integration, and full light/dark theming. This plan delivers that as a single coherent visual + capability uplift on top of the existing architecture — no rewrite, all additive.

Goals from clarification:

- Android features: **Local med-reminder notifications, Haptics + Material You dynamic color, Home-screen widget for vitals**
- Visual: **Material You dynamic + full dark mode** (enhance both light & dark)
- Animation: **Lottie + shared-element transitions + gesture interactions**

Workstreams

1. Theme system overhaul — dynamic color + dark mode

Files: `app/src/main/java/com/carepulse/app/ui/theme/Color.kt`, `Theme.kt`, `Type.kt`

- Define complete **light + dark ColorSchemes** (primary/secondary/tertiary, surface tiers, container variants, on-* roles) using the existing Teal/Navy as the brand fallback when dynamic color is unavailable.
- Use `dynamicLightColorScheme(context)` / `dynamicDarkColorScheme(context)` on **Android 12+** (`Build.VERSION.SDK_INT >= S`); fall back to the hand-tuned brand schemes below.
- Add elevated **surface tiers** (`surfaceContainerLow/High/Highest`) and tonal elevation for the modern depth look.
- Add a user **theme-mode preference** (System / Light / Dark) persisted via DataStore; wire into `CarePulseTheme` composable.
- Refresh typography with a slightly larger display scale and tighter letter-spacing for a modern feel.
- Update `Shapes` to add `extraLarge = 28.dp` for hero cards.

2. Refreshed button + component palette

Files: `app/src/main/java/com/carepulse/app/ui/components/CommonComponents.kt`

- **PrimaryButton**: switch from flat fill to **gradient (primary → tertiary)** with subtle inner shadow; keep spring-press scale; add **haptic** (`HapticFeedbackType.LongPress`) on press.
- Add **SecondaryButton** (tonal), **GhostButton** (outline), **DestructiveButton** (error container) variants so screens stop reusing PrimaryButton for everything.
- **PastelCard** → **ElevatedSurfaceCard** using `surfaceContainerHigh` + 1dp tonal elevation; rounded `extraLarge`.
- Add a **GradientHeader** component for dashboard tops (primary→tertiary gradient with curved bottom).
- Audit every screen for hard-coded `Color(0xFF...)` and replace with `MaterialTheme.colorScheme.*` so dark mode + dynamic color flow everywhere.

3. Animation system upgrade

Dependencies to add (`app/build.gradle.kts`):

- `com.airbnb.android:lottie-compose:6.4.0`
- `androidx.compose.animation:animation-graphics` (for `AnimatedVectorDrawable`)
- Ensure `androidx.compose.animation` is on a version supporting `SharedTransitionLayout` (Compose BOM 2024.06+).

Implementations:

- **Lottie**: empty-state animations (no messages, no vitals yet, no bookings), success confetti on shift end, loading heartbeat on `PulseDashboard`. Store JSON in `app/src/main/res/raw/`.
- **Shared-element transitions**: wrap `CarePulseNavGraph` in `SharedTransitionLayout`; pair caregiver photo + name across `CustomerDashboard` ↔ `CaregiverDetail`, and vitals card ↔ `VitalsLogScreen`.
- **Gestures**:
 - Swipe-to-complete on shift list items (`SwipeToDismissBox` with custom background).
 - Pull-to-refresh on dashboards (`PullToRefreshBox`).
 - Drag-to-dismiss on bottom-sheet flows.
- **Micro-interactions**: ripple + haptic on bottom-nav tab change, animated count-up reuse for new stat tiles, animated `FloatingActionButton` morph on Log Vitals.
- Standardize motion tokens in a new `ui/theme/Motion.kt`: durations (short/medium/long) + easing curves matching Material 3 motion spec.

4. Android platform features

4a. Local med-reminder notifications

- New package `data/reminders/` with `MedicationReminder` model and Room (or `DataStore`) persistence.

- `ReminderScheduler` using `AlarmManager` + `setExactAndAllowWhileIdle` for time-of-day reminders; reschedule on `BOOT_COMPLETED`.
- `ReminderReceiver` (`BroadcastReceiver`) posts a notification on a new `medication_reminders NotificationChannel` (`IMPORTANCE_HIGH`).
- New screen `ui/screens/reminders/RemindersScreen.kt` (list + add/edit) reachable from Caregiver tab or Settings.
- Handle `POST_NOTIFICATIONS` runtime permission (Android 13+) and `SCHEDULE_EXACT_ALARM` (Android 12+).

4b. Haptics + Material You

- Dynamic color already covered in Workstream 1.
- Centralize haptic calls in a `Haptics` util (`performTap`, `performSuccess`, `performError`) wrapping `LocalHapticFeedback` + `Vibrator` for richer patterns on Android 12+ (`VibrationEffect.createPredefined`).
- Apply to: button press, nav tab switch, swipe-to-complete success, vitals submit, error toasts.

4c. Home-screen widget for vitals

- New module-internal package `widget/`:
 - `VitalsWidget` using **Jetpack Glance** (`androidx.glance:glance-appwidget`).
 - Displays last-logged vitals (HR, BP, mood) + upcoming shift; tap opens app to `PulseDashboard`.
 - Update via `GlanceAppWidgetReceiver` triggered after `logVitals` succeeds (call `VitalsWidget.update(context)` from repo).
- Add `widget-info.xml`, manifest receiver entry, preview image.

5. Manifest, permissions, dependencies

- `AndroidManifest.xml`: add `POST_NOTIFICATIONS`, `SCHEDULE_EXACT_ALARM`, `USE_EXACT_ALARM`, `RECEIVE_BOOT_COMPLETED`, `VIBRATE`; register `ReminderReceiver`, `BootReceiver`, `GlanceAppWidgetReceiver`.
- `build.gradle.kts`: add Lottie, Glance (`glance-appwidget`, `glance-material3`), `DataStore` (`datastore-preferences`), `Room` (if chosen for reminders).
- Keep `minSdk = 24`; guard dynamic color, exact-alarm, and notification-permission code with SDK checks.

Critical Files (modify / create)

Modify

- `ui/theme/Color.kt`, `Theme.kt`, `Type.kt`
- `ui/components/CommonComponents.kt`

- `navigation/CarePulseNavGraph.kt` (wrap in `SharedTransitionLayout`, motion tokens)
- `app/build.gradle.kts`, `AndroidManifest.xml`
- `data/repository/FirestoreCarePulseRepository.kt` (trigger widget update after `logVitals`)
- All screen files using hard-coded colors (audit pattern — replace `Color(0xFF...)` with `MaterialTheme.colorScheme.*`)

Create

- `ui/theme/Motion.kt`, `ui/theme/ThemePreference.kt`
- `ui/components/GradientHeader.kt`, `ui/components/buttons/ButtonVariants.kt`
- `data/reminders/MedicationReminder.kt`, `ReminderScheduler.kt`, `ReminderReceiver.kt`, `BootReceiver.kt`
- `ui/screens/reminders/RemindersScreen.kt`
- `util/Haptics.kt`
- `widget/VitalsWidget.kt`, `widget/VitalsWidgetReceiver.kt`, `res/xml/vitals_widget_info.xml`
- `res/raw/lottie_*.json` (empty-states, success, loading)

Reuse (do NOT recreate)

- `PrimaryButton` press-scale animation pattern — extend, don't replace.
- `slideInVertically` staggered card entrance in dashboards — keep, layer shared-element on top.
- `callbackFlow` + `StateFlow` repo pattern — reminders/widget data sources follow the same shape.
- Existing shimmer skeleton — keep for list loaders.

Verification

1. **Build & install** on an Android 12+ device and an Android 10 device (dynamic-color fallback).
2. **Theme:** toggle System/Light/Dark in Settings; change wallpaper on Android 12+ and confirm palette shifts; verify no hard-coded color leaks in dark mode (screen-by-screen pass).
3. **Buttons:** every screen — primary/secondary/ghost/destructive render correctly in both themes; haptic fires on press.
4. **Animations:**
 - Navigate Customer → CaregiverDetail and confirm shared-element transition.
 - Pull-to-refresh on each dashboard.
 - Swipe a shift item to complete; confirm haptic + animated dismissal.
 - Trigger empty states (new account) and confirm Lottie plays.

5. **Reminders:** create a reminder 2 min out, lock device, confirm notification fires; reboot device, confirm reminder still scheduled.
6. **Widget:** add Vitals widget to home screen; log new vitals in-app; confirm widget refreshes within a few seconds; tap widget → opens PulseDashboard.
7. **Permissions:** fresh install on Android 13+ prompts for `POST_NOTIFICATIONS`; denial path handled gracefully.
8. **Regression:** existing Firestore chat, booking, vitals log flows still work end-to-end.